

CY 514: Chemical and Statistical Thermodynamics

Course Code	CY 514
Course Name	Chemical and Statistical Thermodynamics
L-T-P-C	3-0-0-3
Prerequisites	Undergraduate-Level Physical Chemistry or Teacher's Consent
Intended for	BS Chemical Sciences and MSc Chemistry
Distribution	Core
Mutual Exclusion	-
Approval	57th BoA (Oct 2024), in the M.Sc Chemistry curriculum
Revised	59th BoA (Apr 2025) - syllabus revised

Course Contents

- **The Laws of Thermodynamics:** Zeroth law of Thermodynamics, Equilibrium, State Functions, Probability and distribution, Chemical Systems and Surroundings, temperature, equations of state First Law of thermodynamics: Internal energy, heat capacity, enthalpy, Isothermal, Adiabatic and Isobaric Processes, Energy, Enthalpy and Exact Differentials, Heat Capacities, Joule Thompson Effect, Heat Engines and Heat Pumps Second Law of thermodynamics: Entropy, Carnot cycles, heat engines, spontaneous changes, enthalpy and surrounding, Gibbs energy and application of Gibbs Energy Third Law of Thermodynamics: Concept of the absolute zero temperature. [14 Lectures]
- **Statistical Thermodynamics:** Kinetic theory of gases, Probability and Maxwell Boltzmann distribution, Molecular Partition Functions, Thermodynamics from partition Functions, Equilibrium Constants, canonical ensemble; ideal monoatomic, diatomic and polyatomic gases, quantum statistics Electronic, Vibrational, Rotational Partition Functions, Translational Partition Functions, Heat Capacities, Heat Capacities of Solids, Debye and Einstein Models. [14 Lectures]
- **Chemical Equilibria and Reaction Thermodynamics:** Equilibrium: Free Energy and Equilibria, Application of Gibbs energy for Phase change, Helmholtz Free Energies, Gibbs Helmholtz Equation, Free Energies of Formation, phase rule, Clapeyron equation, phase diagram, Ideal and non-ideal solutions, gases, liquids and solutions, equation of states, Fugacities and their determination, Entropy and Free Energy of Mixing, Partial Molal Quantities and the Chemical Potential, Activities and Activity Coefficients, Debye Hückel Theory and Extensions, the Nernst equation, colligative properties, multicomponent phase diagram, Determination of Activity Coefficients, Thermodynamics of Surface and Catalytic Reactions: Adsorption Thermodynamics (Langmuir and BET Isotherms), Heterogeneous Equilibria and Surface Reactions, Thermodynamics of Catalysis and Enzyme Kinetics, Gibbs Energy Minimization in Industrial Processes. [14 Lectures]

References: (Books and articles)

- Donald A Mcquarrie and Simon, **Molecular Thermodynamics**, Viva Student Edition, 2010.
- Glasstone, **Chemical Thermodynamics**, Lightning Source Inc., 2007.
- Thomas Engel and Philips Reid, **Physical Chemistry**, 3rd Edition, Pearson Education, 2013.
- Berry Rice and Ross, **Physical Chemistry books**, 2nd Edition, OUP, 2000.
- R.P. Rastogi and R.R. Misra, **An Introduction to Chemical Thermodynamics**, 2000.